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Roll No

EC-224-CBCS

B.E. III Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Measurements and Instrumentation

Time : Three Hours

Maximum Marks : 60

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Sketch neat diagram, if any.

1. a) Explain the principle of extending the range of AC ammeter.
b) Define the following terms:
 - i) Accuracy and Precision
 - ii) Sensitivity
 - iii) Bolometer
 - iv) Hysteresis
2. a) Draw the block diagram of a CRO and explain briefly its vertical deflection system.
b) Explain the measurement of inductance and capacitance using Maxwell's bridge.
3. a) What is Linear variable Differential Transformer? Explain its working.
b) Explain strain gauge with its derivation.

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4.
 - a) Explain working of sweep frequency generator and pulse wave generator.
 - b) Write short notes on :
 - i) Beat frequency oscillator
 - ii) LCD
5.
 - a) Compare analog and digital instruments.
 - b) Describe the working Digital voltmeters and multimeter.
6.
 - a) Define rectifiers in terms of AC voltmeter and chopper type DC voltmeter.
 - b) Describe different types of sweeps used in CRO.
7.
 - a) Explain the working of Resistance Temperature Detector (RTD) and Thermistor.
 - b) Explain the working of dual slope integrating type ADC with diagram.
8. Write short notes on the following:
 - i) Piezo-electric transducer
 - ii) Loading Effect
 - iii) Optical transducers

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